



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Applied DevOps

Experiment 8

MTECH - School of Computer Science
CYBER SECURITY AND FORENSICS

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Objective — To design and implement a CI/CD pipeline using Docker and Jenkins that automates application build, testing, and deployment across multiple environments, ensuring reliability and deployment consistency.

Task to be Performed

Task 1: Application Setup

Step 1: Create Directory

```
cicd-exp8 — -zsh — 122x34

Last login: Thu Apr  2 11:18:31 on ttys000
[sheoraninfosec@Jigeshs-MacBook-Air ~ % pwd
/Users/sheoraninfosec
[sheoraninfosec@Jigeshs-MacBook-Air ~ % mkdir cicd-exp8
[sheoraninfosec@Jigeshs-MacBook-Air ~ % cd cicd-exp8
sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % nano index.html
```

Step 2: Create Web App

```
cicd-exp8 — nano index.html — 122x34
UW PICO 5.09 File: index.html

<!DOCTYPE html>
<html>
<head>
  <title>CI/CD Pipeline</title>
</head>
<body>
  <h1>Dev Environment for Exp 8</h1>
  <p>Automated deployment using Jenkins CI/CD pipeline.</p>
</body>
</html>
```

Task 2: Dockerfile (Infrastructure as Code)

```
sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % nano Dockerfile

cicd-exp8 — nano Dockerfile — 122x34
UW PICO 5.09 File: Dockerfile

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

Nginx : Docker pulls a ready-made web server so you don't have to install anything manually (no apt install, no setup)

latest is risky in production (can break builds)

Task 3: Github Integration

```
cicd-exp8 — zsh — 122x34

[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/sheoraninfosec/cicd-exp8/.git/
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git add .
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git commit -m "1st commit"
[master (root-commit) d3f3aea] 1st commit
 2 files changed, 15 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.html
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git remote add origin git@github.com:sheoraninfosec/cicd-exp8.git
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git branch -M main
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git push -u origin main
fatal: 'git@github.com:sheoraninfosec/cicd-exp8.git' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 %
```

Forgot to create the git remote repo.

Quick setup — if you've done this kind of thing before

 Set up in Desktop

 or

HTTPS

SSH

git@github.com:sheoraninfosec/cicd-exp8.git



Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

Now, as i had already added the origin url for a repo which didn't existed on remote server so i need to set origin again to the correct url.

Using this command:

git remote set-url origin git@github.com:sheoraninfosec/cicd-exp8.git

```
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git remote set-url origin git@github.com:sheoraninfosec/cicd-exp8.git
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 463 bytes | 463.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/cicd-exp8.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 %
```

Task 4: Jenkins Setup

1. Set up a pipeline job in Jenkins.
2. Configure the pipeline to connect to the GitHub repository.

```
pipeline {
  agent any

  stages {
    stage('Build Docker Image') {
      steps {
        sh 'docker build -t cicd-exp8 .'
      }
    }

    stage('Test Application') {
      steps {
        sh '''
          docker rm -f test-container || true
          docker run -d -p 8081:80 --name test-container cicd-exp8
          sleep 5
          curl -f http://localhost:8081
          docker stop test-container
          docker rm test-container
          '''
      }
    }

    stage('Deploy to Dev') {
      steps {
        sh '''
          docker rm -f dev-container || true
          docker run -d -p 8080:80 --name dev-container cicd-exp8
          '''
      }
    }

    stage('Deploy to Prod') {
      steps {
        sh '''
          docker rm -f prod-container || true
          docker run -d -p 9090:80 --name prod-container cicd-exp8
          '''
      }
    }
  }
}
```

```
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git add Jenkinsfile
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git commit -m "Added Jenkins pipeline again"
[main 5a4c1eb] Added Jenkins pipeline again
1 file changed, 42 insertions(+)
create mode 100644 Jenkinsfile
[sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 % git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 602 bytes | 602.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/cicd-exp8.git
   d3f3aea..5a4c1eb  main -> main
sheoraninfosec@Jigeshs-MacBook-Air cicd-exp8 %
```

Configuration:

- Definition → Pipeline script from SCM
- SCM → Git

Enter your repo: <https://github.com/sheoraninfosec/cicd-exp8.git>

Branch: **main**

Script Path: **Jenkinsfile**

Task 5: Build Stage

1. Configure the pipeline to: Clone the repository
2. Build a Docker image from the Dockerfile

Builds > ⋮

Today

⋮ #1 9:45 pm 🔄 ✖ ⌵

Today

✓ #2 9:47 pm

⊗ #1 9:45 pm

Console Output

Console

 Download

 Copy

View as plain text

```
Started by user dwc
Obtained Jenkinsfile from git https://github.com/sheoraninfosec/cicd-exp8.git
[Pipeline] Start of Pipeline
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Dev Output

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☁	cicd-exp8	2 min 19 sec #2	4 min 8 sec #1	7.2 sec